

Advanced Computer Networks



Dr Sudipta Saha

Associate Professor

Dept of Computer Science & Engineering

Indian Institute of Technology Bhuvaneshwar

Tutorials on CSMA



**DSSRG: Decentralized
Smart Systems Research
Group**

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Question 1.1

Consider a wireless network where multiple peers share a common communication channel.

The following parameters are given:

- Average distance between any two peers = **1 km**
- Signal propagation speed = **2×10^8 m/s**
- Average packet length = **1000 bits**
- Transmission rate = **1 Mbps**

Determine whether:

- **ALOHA (or Slotted ALOHA)** is more suitable, or
- **CSMA-based protocols** are more efficient.

Justify your answer.



Answer 1.1

Transmission time of one packet:

$$T_t = \frac{1000 \text{ bits}}{10^6 \text{ bits/s}} = 10^{-3} \text{ s} = 1 \text{ ms}$$

Propagation delay over 1 km:

$$T_p = \frac{1000}{2 \times 10^8} = 5 \times 10^{-6} \text{ s} = 5 \mu\text{s}$$

Now compare them:

$$a = \frac{T_p}{T_t} = \frac{5 \mu\text{s}}{1000 \mu\text{s}} = 0.005$$



Answer 1.1

This is **very small**. That means a station can sense the channel and learn its state much faster than the time needed to transmit a packet. So carrier sensing is very useful: most collisions can be avoided, and if a collision occurs, only a small amount of time is wasted.

By contrast:

- Pure ALOHA max throughput $\approx 18.4\%$
- Slotted ALOHA max throughput $\approx 36.8\%$

These are much lower because ALOHA does not listen before transmitting, so collisions are frequent.

Since here propagation delay is negligible compared with packet transmission time, CSMA protocols will significantly outperform ALOHA / Slotted ALOHA.

Conclusion: Use CSMA-based protocols. They are better suited and more efficient for this network.



Question 1.2

Consider a wireless network where multiple peers share a common communication channel.

The following parameters are given:

- Average distance between any two peers = **200 km**
- Signal propagation speed = **2×10^8 m/s**
- Average packet length = **1000 bits**
- Transmission rate = **1 Mbps**

Determine whether:

- **ALOHA (or Slotted ALOHA)** is more suitable, or
- **CSMA-based protocols** are more efficient.

Justify your answer.



Answer 1.2

Packet transmission time:

$$T_t = \frac{1000 \text{ bits}}{10^6 \text{ bits/s}} = 10^{-3} \text{ s} = 1 \text{ ms}$$

Propagation delay over 200 km:

$$T_p = \frac{200\,000}{2 \times 10^8} = 10^{-3} \text{ s} = 1 \text{ ms}$$

So,

$$a = \frac{T_p}{T_t} = \frac{1 \text{ ms}}{1 \text{ ms}} = 1$$



Answer 1.2

This is the critical point: **propagation delay is as large as packet transmission time.**

That means in CSMA:

- a node may sense the channel idle,
- but before the information about another node's transmission reaches it, it may already start transmitting,
- so collisions are still very likely.

Thus, the main advantage of CSMA — **listening before transmitting** — becomes much less useful when T_p is comparable to T_t .



Answer 1.2

Comparison

- CSMA works best when $T_p \ll T_t$
- ALOHA / Slotted ALOHA become more competitive when $T_p \approx T_t$ or larger

Since here:

$$T_p \approx T_t$$

CSMA loses much of its benefit, and a simpler random-access scheme like **Slotted ALOHA** is more suitable.



Question 1.3

Consider a wireless network where multiple peers share a common communication channel.

The following parameters are given:

- Average distance between any two peers = **100 m**
- Signal propagation speed = **2×10^8 m/s**
- Average packet length = **1000 bits**
- Transmission rate = **1 Mbps**

Determine whether:

- **ALOHA (or Slotted ALOHA)** is more suitable, or
- **CSMA-based protocols** are more efficient.

Justify your answer.



Answer 1.3

Transmission time:

$$T_t = \frac{1000 \text{ bits}}{10^6 \text{ bits/s}} = 10^{-3} \text{ s} = 1 \text{ ms}$$

Propagation delay over 100 m:

$$T_p = \frac{100}{2 \times 10^8} = 5 \times 10^{-7} \text{ s} = 0.5 \mu\text{s}$$

Now compare:

$$a = \frac{T_p}{T_t} = \frac{0.5 \mu\text{s}}{1000 \mu\text{s}} = 0.0005$$

This is extremely small.



Question 2.1

Consider a network of **10 nodes** using **p-persistent CSMA** over a shared channel.

At a given time:

- All nodes have packets to transmit (these are called **active nodes**).
- All active nodes attempt transmission using the same protocol.
- Assume that all active nodes have equal priority and identical traffic conditions.

In p-persistent CSMA, what should be the value of p ?



Answer 2.1

Given:

- Number of active nodes = $N = 10$

Each node transmits with probability p , so the **expected number of transmitting nodes** is:

$$N \cdot p$$

To minimize collisions, we want:

$$N \cdot p \approx 1$$

$$10 \cdot p = 1 \Rightarrow p = \frac{1}{10} = 0.1$$

$$\boxed{p \approx 0.1}$$



Question 2.2 [for discussion]

Consider a network of **10 nodes** using **p-persistent CSMA** over a shared channel.

At a given time:

- Only a subset of nodes have packets to transmit (these are called **active nodes**).
- All active nodes attempt transmission using the same protocol.
- Assume that all active nodes have equal priority and identical traffic conditions.

Suppose different nodes are assigned **different values of p** based on their activity level (e.g., more active nodes use higher p , less active nodes use lower p):

- What impact will this have on:
 - collision probability
 - fairness
 - channel utilization?



Answer 2.2

Assigning **different p** values based on activity level changes the behavior in three ways:

Collision probability

It will usually **increase**, especially if the more active nodes are given **higher p** .

Why:

- In p -persistent CSMA, when the channel is idle, each active node transmits with probability p .
- If several highly active nodes all use larger p , they are more likely to transmit in the same slot.
- That raises the chance of **simultaneous attempts**, hence more collisions.

So unequal p values, particularly with aggressive nodes using high p , tend to make collisions more likely than a well-chosen common p .

Fairness

It becomes **worse**.

Why:

- Nodes with higher p access the channel more aggressively.
- Nodes with lower p defer more often.
- Over time, high- p nodes get a larger share of transmissions, while low- p nodes may experience long delays or partial starvation.

So fairness is reduced because access is no longer symmetric.



Answer 2.2

Channel utilization

It can go **either way**, but in general it becomes **less balanced** and may degrade if the p values are not chosen carefully.

Two effects compete:

- If all p values are too low: the channel stays idle too often, so utilization drops.
- If some p values are too high: collisions increase, wasting channel time, so utilization also drops.

A small amount of differentiation might help in some engineered priority system, but under your assumption of equal-priority nodes, using different p values usually gives:

- **more collisions**
- **less fairness**
- **worse or less stable utilization overall**

Conclusion

If more active nodes use higher p and less active nodes use lower p , then:

- **Collision probability:** generally **increases**
- **Fairness:** **decreases**
- **Channel utilization:** may become inefficient; often **degrades overall** unless very carefully tuned



Question 3

Consider a CSMA-based network.

Given

- Mean distance between any two peers = **2 km**
- Propagation speed = **2×10^8 m/s**
- Transmission rate = **1 Mbps**
- Offered Load, $G = 1$

Case 1 (Small Packet) Packet size = **500 bits**

Case 2 (Medium Packet) Packet size = **5000 bits**

Case 3 (Large Packet) Packet size = **20,000 bits**

For each case, compute Throughput

Denote:

- $S_1 \rightarrow$ Case 1
- $S_2 \rightarrow$ Case 2
- $S_3 \rightarrow$ Case 3

Compare S_1, S_2, S_3



Answer 3

Show the graphs...



Question 4.1

Consider a shared wireless network with **100 nodes** using a CSMA-based protocol. At any given time, only a subset of nodes have packets to transmit (active nodes).

Assume:

- All nodes can sense the channel perfectly.
- Propagation delay is negligible.
- All active nodes have equal priority.

80 nodes are active simultaneously

Determine which protocol is more suitable:

- 1-persistent CSMA
- 0-persistent CSMA
- p-persistent CSMA

If p-persistent CSMA is preferred, suggest an appropriate value of p .



Answer 4.1

Let the formulas correspond to:

1. 1-persistent CSMA

$$S_1 = \frac{G[1 + G + aG(1 + G + aG/2)]e^{-G(1+2a)}}{G(1 + 2a)(1 + aG) - (1 - e^{-aG}) + aGe^{-G(1+2a)}}$$

2. Non-persistent CSMA (your "0-persistent")

$$S_0 = \frac{Ge^{-aG}}{G(1 + 2a) + e^{-aG}}$$

3. p-persistent CSMA

$$S_p = \frac{(a + p)Ge^{-(a+p)G} - pGe^{-(2a+p)G}}{1 + a - e^{-aG}}$$



Answer 4.1

Since propagation delay is negligible

That means:

$$a \approx 0$$

and with 80 active nodes, the offered load G is large.

1) 1-persistent CSMA

Putting $a = 0$ in formula 1:

$$S_1 \approx (1 + G)e^{-G}$$

For $G = 80$:

$$S_1 \approx 81e^{-80} \approx 0$$

So throughput is almost zero.



Answer 4.1

2) Non-persistent CSMA

Putting $a = 0$ in formula 2:

$$S_0 \approx \frac{G}{G + 1}$$

For $G = 80$:

$$S_0 \approx \frac{80}{81} \approx 0.988$$

So throughput is **very high**.



Answer 4.1

3) p-persistent CSMA

For the third formula, if you put $a = 0$ exactly, the denominator becomes problematic, so we evaluate it for **very small** a rather than exactly zero.

That means:

- p-persistent can work well,
- but its throughput depends strongly on the choice of p ,
- and for heavy load $G = 80$, it needs a **very small** p .

Using a practical choice like

$$p \approx \frac{1}{80} = 0.0125$$

the throughput is still typically **below non-persistent CSMA** for very small a .

So from these formulas, **p-persistent** is **better** than **1-persistent**, but **non-persistent** gives the **highest throughput** under heavy contention and negligible propagation delay.



Question 4.2

Consider a shared wireless network with **100 nodes** using a CSMA-based protocol. At any given time, only a subset of nodes have packets to transmit (active nodes).

Assume:

- All nodes can sense the channel perfectly.
- Propagation delay is negligible.
- All active nodes have equal priority.

30 nodes are active simultaneously

Determine which protocol is more suitable:

- 1-persistent CSMA
- 0-persistent CSMA
- p-persistent CSMA

If p-persistent CSMA is preferred, suggest an appropriate value of p.



Answer 4.2

p-persistent CSMA is the most suitable choice.

Why:

- **1-persistent CSMA** is too aggressive. With **30 active nodes**, when the channel becomes idle, many nodes may transmit immediately, causing a high chance of collision.
- **0-persistent CSMA** reduces collisions, but it is too conservative because nodes back off randomly and may leave the channel idle unnecessarily.
- **p-persistent CSMA** gives the best balance: it reduces collisions while still keeping the channel well utilized.

Choosing p

A common rule is to choose:

$$p \approx (1 / \text{number of active nodes})$$

Here, $p \approx (1 / 30) = 0.033$

So an appropriate value is: $p \approx 0.03$



Question 4.3

Consider a shared wireless network with **100 nodes** using a CSMA-based protocol. At any given time, only a subset of nodes have packets to transmit (active nodes).

Assume:

- All nodes can sense the channel perfectly.
- Propagation delay is negligible.
- All active nodes have equal priority.

2 nodes are active simultaneously

Determine which protocol is more suitable:

- 1-persistent CSMA
- 0-persistent CSMA
- p-persistent CSMA

If p-persistent CSMA is preferred, suggest an appropriate value of p.



Answer 4.3

1-persistent CSMA

- When the channel becomes idle, a node transmits immediately.
- With only 2 active nodes, collision risk is relatively low.
- This gives **high channel utilization** and **low delay**.

0-persistent (non-persistent) CSMA

- Nodes wait random times before retrying if the channel is busy.
- This reduces collisions, but with only 2 nodes that benefit is small.
- It may waste time by leaving the channel idle unnecessarily.

p-persistent CSMA

- Also workable, but mainly useful when contention is moderate or high.
- With only 2 nodes, setting $p < 1$ just adds unnecessary waiting.
- If used, you would choose a fairly large p , around

$$p \approx (1/2) = 0.5$$

but even then, 1-persistent is usually simpler and more efficient here.

Conclusion

The best choice is: 1-persistent CSMA



Question 5

Consider a network using **0-persistent CSMA**. Over a long observation interval, the channel behavior is measured as follows:

- The channel is **idle 40% of the time**.
- The channel is **busy 60% of the time**.
- Out of the busy time, only **30% corresponds to successful data transmission** (the remaining busy time is due to collisions and retransmissions).

Compute the **throughput S** of the system.



Answer 5

Given:

- Idle time = 40%
- Busy time = 60%
- Out of busy time:
 - Successful transmission = 30%
 - Remaining = collisions/retransmissions

Step 1: Successful fraction of total time

Only **30% of the busy time** is successful:

$S = (\text{busy fraction}) \times (\text{successful fraction of busy time})$

$S = 0.6 \times 0.3 = 0.18$

Final Answer:

$S = 0.18$ (or 18%)



Question 6

Consider a network using **1-persistent CSMA**. Over a long observation interval, the channel behavior is measured as follows:

- The channel is **idle 20% of the time**.
- The channel is **busy 80% of the time**.
- Out of the busy time:
 - **50% corresponds to successful transmissions**
 - The remaining busy time is due to collisions.

Compute the **throughput S** of the system.



Answer 6

Throughput S is the **fraction of total time used for successful transmissions**.

Given:

- Idle time = 20% $\rightarrow$ 0.2
- Busy time = 80% $\rightarrow$ 0.8
- Out of busy time:
 - Successful = 50% $\rightarrow$ 0.5

Step 1: Compute throughput

$S = (\text{busy fraction}) \times (\text{successful fraction of busy time})$

$S = 0.8 \times 0.5 = 0.4$

Final Answer:

$S = 0.4$ (or 40%)

